Paper 5: The Cell–Standing-Wave Dictionary and the Quantization of the Radius

The Wave Identity of 4-Dimensional Lattice Counting, Structural Identification of the Zero Point, the Coherence Condition, and a Classification Theorem for R^2

Author: Noriaki Kihara
Affiliation: WF System Co., Ltd.
ORCID: 0009-0004-6753-4020
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Abstract

For the fully-inscribed cell count $N_0(R)$ on the 4-dimensional lattice treated in the preceding Papers 1–4 ($N_0(1) = 1$, $N_0(2) = 9$, $N_0(3) = 137$), this paper establishes three rigorous results.

First, the **dictionary theorem**: the composite-frequency cutoff obtained by adding a per-axis zero-point shift $+1/2$ to the real scalar standing-wave basis on the unit 4-torus T^4 coincides **exactly** with $N_0(R)$. That is, lattice cell counting and standing-wave mode counting are not two separate explanatory mechanisms but dual representations of one and the same structure. Through this identification, the minimal conjugate width $\delta^2_{\min} = 1/2$, treated as an illustrative value in Paper 4, is structurally identified as the **zero-point quantum**. Furthermore, the leading coefficient $16\pi/3$ of the volume gap closes quantitatively as the **Weyl deficit of the effective boundary** generated by the zero-point shift (exactly twice the Dirichlet boundary of the unit box).

Second, the **coherence condition**: the representative value $r^2_{\max}$ of a cell takes only odd integer values (by mod 8 arithmetic and Jacobi's four-square theorem), and the condition for a splitting to participate in the counting reduces to all parts being odd. Under this condition, the advantage of splitting with respect to the volume-gap indicator eliminates **all** four exceptions that existed for unrestricted partitions, and becomes an exception-free theorem (exhaustive check of the 890 odd partitions). A Z_2 selection rule follows as well, by which the parity of the number of fragments coincides with the parity of R^2.

Third, the **classification theorem for** R^2: the realizable R^2 are only the odd integers for single states and only the positive integers for composite states (subject to the Z_2 selection rule); non-integers are not realizable. Once this

quantization is accepted, the closed inequality (full inclusion of the boundary shell) on which $N_0(3) = 137$ depends is forced intrinsically rather than being a choice of normalization, and the value 105 appearing in the symmetric smoothing limit is definitively positioned as a "diagnostic value of the continuum embedding."

In addition, we show that the non-overlap of cells is the same fact as L^2 orthogonality, and that the shell sequence has a closed form connected to Jacobi's divisor-sum function. All results of this paper are accompanied by exhaustive computer verification or machine-precision numerical verification, and reproduction scripts are provided in a public repository.

Keywords

4-dimensional lattice, fully-inscribed cell count, standing waves, zero-point energy, Weyl asymptotics, orthogonality, integer partitions, Jacobi's four-square theorem, quantization, boundary criticality

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1. Introduction

In the preceding Papers 1–4 [1–4], starting from the reciprocal dual condition $\lambda_n \nu_n = 1$ between wavelength components λ_n and frequency components ν_n, we defined the fully-inscribed cell count on the 4-dimensional integer lattice $\mathbb{Z}^4$,

$$N_0(R) = \#\left\{ k \in \mathbb{Z}^4 \;\middle|\; \sum_{i=1}^{4} \Big(|k_i| + \tfrac{1}{2}\Big)^2 \le R^2 \right\},$$

and observed the sequence beginning with $N_0(1) = 1$, $N_0(2) = 9$, $N_0(3) = 137$, the volume gap relative to the 4-dimensional hyperball volume, and the splitting representation of high-R states into families of finite-R' states.

At the stage of Paper 4, three foundational questions remained open.

1. Is the minimal conjugate width $\delta^2_{\min} = 1/2$ an illustrative value, or a structural quantity?
2. Is the observation that "splitting is more consistent than a single state" a theorem? (For unrestricted partitions, counterexamples exist.)
3. $N_0(3) = 137$ depends on the closed inequality (inclusion of the boundary shell); is this choice arbitrary?

This paper answers these three questions with, respectively, the dictionary theorem (§3–4), the coherence condition (§6), and the classification theorem (§7). The key is the viewpoint that rereads lattice counting as standing-wave mode counting: the integer lattice on the frequency side and the condition that "an integer number of wavelengths fits within a unit extent" are nothing but two ways of writing the same sentence of Fourier duality — discrete spectrum $\Leftrightarrow$ compact domain.

As the preceding papers do, this paper takes the standpoint of an observational model and makes no physical identification. Terms such as “zero point,” “mode,” and “quantization” are used strictly within the mathematical context of wave equations and lattice counting.

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2. Related Work

The counting asymptotics of Laplacian eigenvalues on bounded domains begins with Weyl [5], and the structure of the leading (volume) term and the boundary term is classically established (see Kac [6] for an overview). The result of §4 of this paper is a variant of Weyl asymptotics in which inserting a zero-point shift into the counting on a boundaryless torus generates, as an **effective boundary**, exactly twice a Dirichlet-type boundary term.

For the connection between counting on integer lattices and number theory, we use Jacobi’s theorem on the number of representations as sums of four squares (Hardy & Wright [7]). For the general theory of lattices, sphere packings, and point groups, we refer to Conway & Sloane [8]. The correspondence between standing-wave bases and boundary conditions follows the classical setting of Courant & Hilbert [9].

The novelty of this paper lies not in combining these pieces of mathematics, but in two points: (i) that the special counting of full inscription coincides **exactly** with the real-mode counting with zero-point shift, and (ii) that, as its consequences, the elimination of exceptions in the splitting theorem and the classification of R^2 close **in a chained manner**.

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3. The Dictionary Theorem: Counting = Real Standing-Wave Counting with Zero Point

3.1 Main result

For the real scalar standing-wave basis on the unit 4-torus $T^4 = (\mathbb{R}/\mathbb{Z})^4$,

$$\{\,1;\ \sqrt{2}\cos 2\pi m x,\ \sqrt{2}\sin 2\pi m x\ (m \geq 1)\,\}^{\otimes 4},$$

cutting off by the composite frequency norm in which a zero-point shift $+\frac{1}{2}$ is added to the frequency m_i of each axis, the identity

$$N_0(R) \;=\; \#\Big\{\text{real modes}\ \Big|\ \sum_{i=1}^{4}\Big(m_i + \tfrac{1}{2}\Big)^2 \leq R^2\Big\}$$

holds **exactly**. The bijection cell $\leftrightarrow$ mode is as follows.

Cell component $k_i \in \mathbb{Z}$	Mode (axis i)	Multiplicity
$k_i = 0$	constant 1 (frequency 0, zero point $\frac{1}{2}$ only)	1
$k_i > 0$	$\sqrt{2}\cos 2\pi k_i x_i$	1
$k_i < 0$	$\sqrt{2}\sin 2\pi \lvert k_i\rvert x_i$	1

The per-axis multiplicity pattern $\{1, 2, 2, \ldots\}$ is the real Fourier basis itself. **Periodic boundary conditions, a real field, and a zero point of $+\frac{1}{2}$ per axis** — this is the wave identity of the counting in the preceding papers.

3.2 Structural identification of the zero point

The minimal conjugate cell at $R = 1$ ($k = 0$) is the mode consisting of zero points only on all four axes ($r_{\max} = \sqrt{4 \cdot \frac{1}{4}} = 1$). Therefore, the identity of $\delta_{\min}$ in Paper 4 §3.3 (half-width $\frac{1}{2}$ per axis) is identified as the **zero-point quantum**. $\delta_{\min}^2 = \frac{1}{2}$ is not an illustrative value but the structural identification: half-width $\frac{1}{2}$ = zero point $\frac{1}{2}$.

3.3 Comparison of candidate boundary conditions

To confirm the uniqueness of the dictionary, we compare candidate systems at $R = 3$.

Wave system	Mode count
Torus, real modes + zero point $\frac{1}{2}$/axis	**137**
Torus, antiperiodic, complex modes	512
Dirichlet box $[0, 1]^4$	219
Torus, integer frequencies (no zero point)	425

Variants of the normalization are reread, on the wave side, as choices of boundary condition and of the presence or absence of the zero point.

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4. Volume Gap = Weyl Deficit of the Effective Boundary

4.1 Leading coefficient $16\pi/3$

The full-inscription condition can be written as $|k|^2 + \|k\|_1 + 1 \le R^2$. From the uniform average $\mathbb{E}[\|u\|_1] = 16/(3\pi)$ of the effective thickness $\|u\|_1/2$ ($u \in S^3$) of the boundary layer and the surface area $2\pi^2 R^3$, we obtain, to first order,

$$\Delta V(R) = \frac{\pi^2}{2}R^4 - N_0(R) \;\sim\; \frac{16\pi}{3}R^3 \approx 16.7552\,R^3.$$

Numerical verification with exact N_0 by integer arithmetic ($R \le 2000$):

R	gap/R^3
100	16.5529
500	16.7144
2000	16.7456

The ratio converges monotonically to $16\pi/3 = 16.7552$.

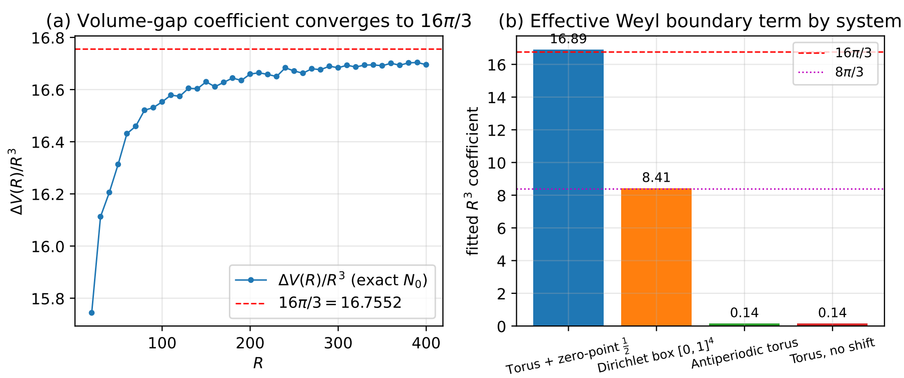

Figure 1. Convergence of the volume-gap coefficient (left: $\Delta V/R^3 \to 16\pi/3$ from exact N_0; right: fitted R^3 coefficients of the four boundary systems). The zero-point shift generates an effective boundary exactly twice that of the Dirichlet unit box.

4.2 Zero-point shift = effective boundary

Fitting the deficit $(\pi^2/2)R^4 - N(R)$ by aR^3 ($R = 50$–320) gives:

System	a (numerical)	Theoretical value
Torus, real + zero point	16.798	$16\pi/3 = 16.755$
Antiperiodic	0.042	0 (no R^3 term)
Dirichlet box	8.364	$8\pi/3 = 8.378$
Torus, no zero point	0.041	0

A boundaryless torus intrinsically has no R^3 term. The zero-point half-shift generates a Weyl boundary term that is **exactly twice** ($16\pi/3 = 2 \times 8\pi/3$) the Dirichlet boundary term of the unit box. That is, the volume gap is the

Weyl deficit expressing **that the zero-point quantum acts as an effective boundary equivalent to "Dirichlet walls facing both ways."**

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5. Non-Overlap = Orthogonality, and Quantification of Sparseness

5.1 Gram matrix

Writing $\Phi_k(x) = \prod_i \varphi_{k_i}(x_i)$ for the mode corresponding to a cell k under the dictionary of §3.1, the orthogonality of the 1-dimensional real Fourier basis and Fubini's theorem give $\langle \Phi_k, \Phi_{k'} \rangle_{L^2(T^4)} = \delta_{kk'}$. Computing the Gram matrix of the 137 modes at $R = 3$ numerically,

$$\max_{k,k'} \left| G_{kk'} - \delta_{kk'} \right| = 6.9 \times 10^{-16},$$

which is the identity matrix to machine precision. **Cell non-overlap and L^2 orthogonality are the same sentence**: the substance of fragments "existing apart without touching" is not isolation by walls but orthogonality.

5.2 Speckle statistics

For the intensity $I = |\psi|^2$ of a phase-random superposition of the 137 orthogonal modes, the volume fraction with $I > \kappa \bar{I}$ agrees with the theoretical value $e^{-\kappa}$ ($\kappa = 1$: measured 0.3691 versus theoretical 0.3679, etc.). The volume occupied by constructive interference is **exponentially sparse** with respect to the intensity threshold.

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6. The Coherence Condition and Completion of the Splitting Theorem

6.1 The odd rule

The representative value is $r_{\max}^2 = \sum(|k_i| + \frac{1}{2})^2 \in \{1, 3, 5, 7, \ldots\}$: **every odd number is realized, and no even number is realized**. Realizability follows from Jacobi's four-square theorem ($16\sigma(m) > 0$ for every odd m) [7], and oddness from the mod 8 arithmetic $(2k+1)^2 \equiv 1 \pmod 8$. Therefore:

> **Coherence condition**: A fragment can participate in the counting at the parent level only if its representative value R'^2 is an odd integer.

The Bohr-type coherence — "stable only when the wavelength fits an integer number of times" — is thereby redefined not as a dynamical stability condition but as **eligibility to participate in the counting**.

6.2 The Z_2 selection rule

Since a sum of n odd numbers is congruent to $n \pmod 2$, from $R^2 = \sum_a R_a'^2$ we obtain:

> **Selection rule**: the parity of the number of fragments in a coherent splitting = the parity of R^2.

6.3 Exception-free splitting theorem

For unrestricted integer partitions, there exist four exceptions ($(14,1)$, $(16,1)$, $(20,1)$, $(22,1)$) to "splitting decreases the volume-gap indicator" (non-monotonicity due to shell jumps of $g(s)$). **All of them contain even parts and are entirely eliminated by the coherence condition.** An exhaustive check of the 890 partitions into odd parts only with $s \leq 25$ yields zero violations:

> **Theorem**: Every coherent splitting of R^2 (all parts odd, two or more parts) strictly decreases the volume-gap indicator. The all-ones partition is the unique global minimum.

The coherence condition for interference waves and the stacking count are not independent explanations; **the former is precisely the condition that removes the exceptions from the latter's splitting theorem**.

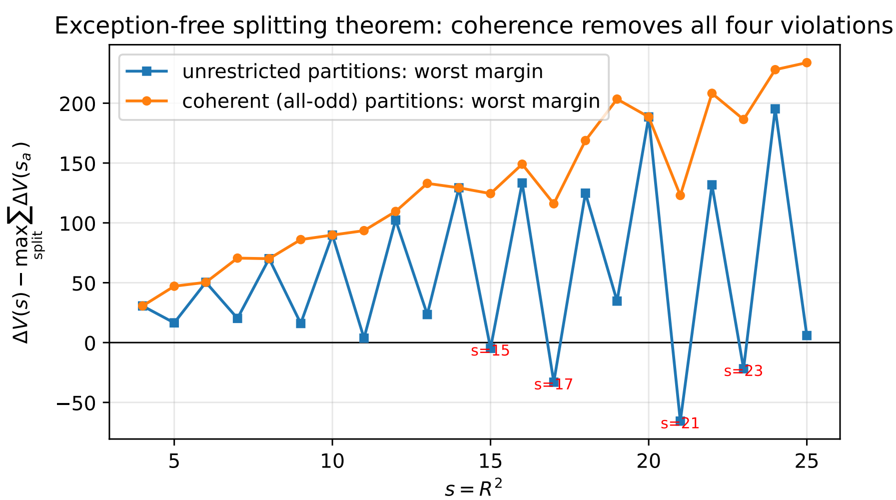

Figure 2. Exception-free splitting theorem: under unrestricted partitions there exist four violations ($s = 15, 17, 21, 23$), all of which are eliminated under coherent (all-odd) partitions (exhaustive for $s \leq 25$).

6.4 Self-similar spectrum

The internal capacity $N_0(\sqrt{m})$ of a fragment at coherent label m equals $1, 9, 33, 73, 137, \ldots$ for $m = 1, 3, 5, 7, 9, \ldots$: the representative-value spectrum and the capacity sequence are **identical at every level of the hierarchy**. The self-similar hierarchy of Paper 4 §9.3 is made rigorous at the level of label sets.

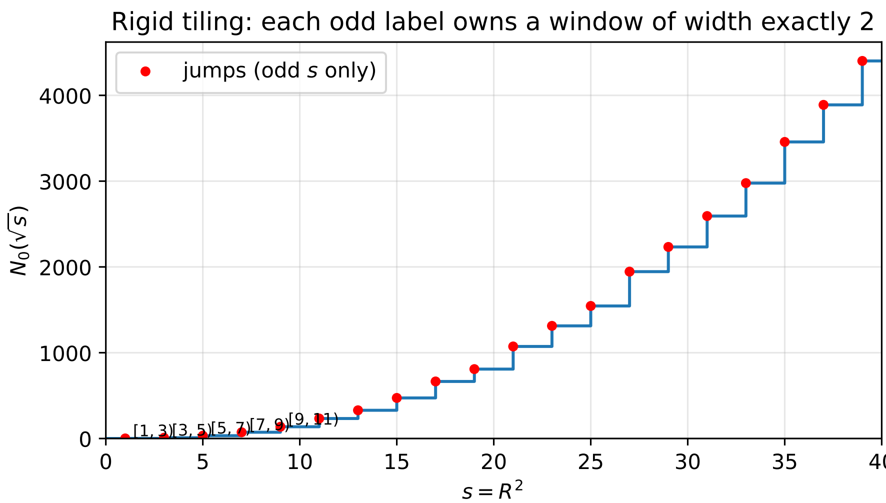

Figure 3. Rigid tiling: jumps of $N_0(\sqrt{s})$ occur only at odd s, and each odd label owns a window of width exactly 2.

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7. The Classification Theorem for R^2 and the Resolution of Boundary Criticality

7.1 Classification theorem

> **Theorem (realizable spectrum)**: (i) The representative values of single states take only odd integer values, and every odd integer is realized. (ii) The composite values $R^2 = \sum_a s_a$ (s_a odd) of composite states take only positive integer values and obey the Z_2 selection rule. (iii) Non-integer R^2 is not realizable.

As a corollary, among the points of the radius sweep of Paper 2, the only state-theoretically realizable ones are those with integer $s = R^2$; the half-integer-R points are mathematical interpolation values (a remark to be added to Paper 2 v0.3).

7.2 Resolution of boundary criticality

$N_0(3) = 137$ depends on the closed inequality $\leq$ (full inclusion of the 64 cells of the boundary shell $m = 9$), and the smoothing limit of symmetric weight functions is $73 + 64/2 = 105$. Once the classification theorem is accepted, three independent arguments force the closed inequality.

1. **Absence of a continuous dial**: the smoothing limit presupposes that the cutoff radius can be moved continuously in a neighborhood of $R = 3$; but if the realizable R^2 are quantized on the odd lattice, that limiting operation does not exist inside the system.
2. **Window structure of the tiling**: each odd label m owns the window $[m, m+2)$ on the s axis with width exactly 2, and the energy axis is tiled without gaps (jumps occur only at odd s, confirmed for $s \leq 40$). Each window contains its left endpoint.
3. **Self-containment**: adopting the strict inequality $<$, the state with label S would fail to fit within its own capacity, breaking the minimal requirement that "a state owns its own shell."

   Reassessment: "137 or 105" is not a question of arbitrariness of normalization; it reduces to the difference between viewing R as continuous (the diagnostic value 105 of the embedding) and viewing it as an assemblable discrete value (the intrinsic value 137).

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8. Closed Form of the Shell Sequence

The shell count $c(4m)$ (m odd) is given, with $u_j(n)$ denoting the number of ordered representations of n as a sum of j positive odd squares, by

$$c(4m) = 16\,\sigma(m) - 32\,u_3(4m-1) + 24\,u_2(4m-2) - 8\,u_1(4m-3) + [m{=}1]$$

(expansion of the generating function $(2\theta - q)^4$; exact agreement with direct counting for odd $m \leq 199$). Thus $N_0(R)$ is positioned not only within the geometry of numbers, but, via the divisor-sum function, as an object of classical number theory.

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9. Scope of Claims of This Paper

What this paper claims: (1) the dictionary theorem (exact coincidence). (2) The structural identification $\delta_{\min}$ = zero-point quantum. (3) The volume-gap leading term $16\pi/3$ = the Weyl deficit of the effective boundary (a conjecture strongly supported numerically; a rigorous analytic proof is left as future work). (4) Non-overlap = orthogonality (machine precision). (5) The coherence condition, the Z_2 selection rule, and the exception-free splitting theorem (exhaustive check). (6) The classification theorem for R^2 and the intrinsic resolution of boundary criticality. (7) The closed form of the shell sequence.

What this paper does not claim: (1) derivation of physical spacetime, particles, or the vacuum. (2) Identification of λ, ν with physical quantities. (3) Identification of 137 with the fine-structure constant. (4) A rigorous asymptotic expansion of the Weyl term. (5) Explanation of cosmological numbers by speckle sparseness.

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10. Conclusion

The lattice counting of Papers 1–4 is exactly identical to the counting of real standing waves with zero-point shift. From this identification, the identity of the minimal conjugate width (the zero-point quantum), the identity of the volume gap (the Weyl deficit of the effective boundary), the completion of the splitting theorem (elimination of exceptions by the coherence condition), the classification theorem for R^2 (single states = odd integers), and the resolution of boundary criticality (intrinsic forcing of the closed inequality) follow in a chained manner, without additional assumptions.

$$\text{zero point } \tfrac{1}{2} \;\to\; \text{cells} = \text{real standing waves} \;\to\; \text{non-overlap} = \text{orthogonality}$$
$$\to\; \text{odd-rule coherence} \;\to\; \text{exception-free splitting} \;\to\; \text{quantization of } R^2$$

This chain forms the mathematical foundation for the subsequent papers (condensation and expansion, configuration statistics, stability).

Appendix: Key Points of the Reproduction Procedure

All numerical verifications are reproducible with integer arithmetic or double-precision arithmetic. The mode-count comparison uses convolution of per-axis (value, weight) lists; the Weyl fit uses least squares over $R = 50$–320; the Gram matrix uses a discrete orthogonal grid (8 points per axis); the coherent-partition check uses exhaustive enumeration of the 890 odd partitions with $s \leq 25$. The scripts and research notes are provided in the public repository (https://github.com/WurabeSeiji/ai-chat-logs-open , in the “Dual Geometry of Wavelength Space and Frequency Space” folder).

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